

Lab taskssss

Task 1

```
#include<iostream>
using namespace std;
struct country
{
    string name;
    float population;
};
int main()
{
    country c;
    cout << "enter country name:\n";
    cin >> c.name;
    cout << "Enter population(in millions):";
    cin >> c.population;

    cout << "\n-----COUNTRY DATA-----\n ";
    cout << "COUNTRY: " << c.name << endl;
    cout << "POPULATION: " << c.population << endl;
    return 0;
}
```

```
C:\Users\Taha's Lappy\OneDrive\Documents\lab 3 lab task 1.cpp.exe
enter country name:
pakistan
Enter population(in millions):4682384691

-----COUNTRY DATA-----
COUNTRY: pakistan
POPULATION: 4.68238e+009

Process exited after 5.82 seconds with return value 0
Press any key to continue . . .
```

Task 2

```
lbtask 2.cpp
#include<iostream>
using namespace std;
struct student
{
    int rollno;
    int marks[5];
};
int main()
{
    student s;
    cout << "enter roll number:\n";
    cin >> s.rollno;
    cout << "enter marks of 5 subjects:\n";
    for(int i=0 ;i<5;i++)
    {
        cin >> s.marks[i];
    }
    cout << "-----STUDENT RECORD-----\n ";
    cout << "roll number: " << s.rollno << endl;
    cout << "marks: ";
    for(int i=0;i<5;i++)
    {
        cout << s.marks[i] << " ";
    }
    return 0;
}
```

```
C:\Users\Taha's Lappy\OneDrive\Documents\lab 3 labtask 2.exe
enter roll number:
12
enter marks of 5 subjects:
33
44
55
66
77

-----STUDENT RECORD-----
roll number: 12
marks: 33 44 55 66 77

Process exited after 3.906 seconds with return value 0
Press any key to continue . . .
```

TASK 3

```
#include<iostream>
using namespace std;
struct car
{
    string model;
    string color;
    float price;
};
int main()
{
    car cars[5];
    cout << "Enter data of 5 cars:\n";
    for(int i=0;i<5;i++)
    {
        cout << "\nCar " << i+1 << endl;
        cout << "Model: ";
        cin >> cars[i].model;
        cout << "Color: ";
        cin >> cars[i].color;
        cout << "Price: ";
        cin >> cars[i].price;
    }
    int choice;
    cin >> choice;
    if(choice==0)
    {
        int maxIndex=0;
        for(int i=1; i<5; i++)
        if(cars[i].price > cars[maxIndex].price)
        maxIndex = i;
        cout << "\n--- Most Expensive Car ---\n";
        cout << "Model: " << cars[maxIndex].model << endl;
        cout << "Color: " << cars[maxIndex].color << endl;
        cout << "Price: " << cars[maxIndex].price << endl;
    }
    else if(choice == 2)
    {
        cout << "\n--- All Cars ---\n";
        for(int i=0; i<5; i++)
        {
            cout << "\nCar " << i+1 << endl;
            cout << "Model: " << cars[i].model << endl;
            cout << "Color: " << cars[i].color << endl;
            cout << "Price: " << cars[i].price << endl;
        }
    }
}
```

C:\Users\Taha's Lappy\OneDrive\Documents\lab 3 labtask 3.cpp.exe

Enter data of 5 cars:

Car 1
Model: i7
Color: pink
Price: 9999999999

Car 2
Model: 2002
Color: blue
Price: 99999998

Car 3
Model: 2010
Color: black
Price: 8678594034708

Car 4
Model: 2000
Color: purple
Price: 888882

Car 5
Model: 2011
Color: red
Price: 444445

TASK 4

```
#include<iostream>
using namespace std;
struct phone
{
    int areacode;
    int exchange;
    int number;
};
int main()
{
    phone mynumber={ 282,141,303};
    phone yournumber;
    cout << "enter your areacode,exchange,number:\n";
    cin >> yournumber.areacode >> yournumber.exchange >> yournumber.number;

    cout << "My NUMBER IS:\n" << mynumber.areacode << " " << mynumber.exchange << " " << mynumber.number << " " << endl;
    cout << "Your number is:\n" << yournumber.areacode << " " << yournumber.exchange << " " << yournumber.number << " " << endl;
    return 0;
}
```

C:\Users\Taha's Lappy\OneDrive\Documents\lab3 labtas...

enter your areacode,exchange,number:

222
3344
55677777

My NUMBER IS:
282 141 303

Your number is:
222 3344 55677777

TASK 5

```
btask 5.cpp
#include<iostream>
using namespace std;
struct points
{
    int x;
    int y;
};
int main()
{
    points p1,p2,p3;
    cout << "enter coordinates for p1: ";
    cin >> p1.x >> p1.y;
    cout << "enter coordinate for p2: ";
    cin >> p2.x >> p2.y;
    p3.x=p1.x + p1.y;
    p3.y=p2.x + p2.y;
    cout<<"Coordinates of p1+p2 are: "<<p3.x<<"", "<<p3.y;
    return 0;
}
```

```
C:\Users\Taha's Lappy\OneDrive\Documents\lab 5 lab task 5.exe
enter coordinates for p1: 4
4
enter coordinate for p2: 3
3
Coordinates of p1+p2 are: 8, 8
-----
Process exited after 7.031 seconds with return value 0
Press any key to continue . . .
```

TASK 6

```
#include<iostream>
using namespace std;
struct Distance
{
    int feet;
    float inches;
};
struct Volume
{
    Distance length;
    Distance width;
    Distance height;
};
float tofeet(Distance d)
{
    return d.feet + (d.inches / 12.0);
}
int main()
{
    Volume room = { {12,6}, {10,0}, {8,3} };

    float l = tofeet(room.length);
    float w = tofeet(room.width);
    float h = tofeet(room.height);

    float volume = l * w * h;

    cout<<"Room volume = "<<volume<<" cubic feet";

    return 0;
}
```

```
C:\Users\Taha's Lappy\OneDrive\Documents\lab 3 lab task 6.exe
Room volume = 1031.25 cubic feet
-----
Process exited after 0.1577 seconds with return value 0
Press any key to continue . . .
```

TASK 7

```
.Users\Taha's Lappy\OneDrive\Documents\lab 3 lab task 6.cpp - [Executing] - Dev-C++ 5.11
Edit Search View Project Execute Tools AStyle Window Help
(globals)
lab task 6.cpp
#include<iostream>
using namespace std;
struct employee
{
    int empnum;
    float sallery;
};
int main()
{
    employee e[3];
    for(int i=0;i<3;i++)
    {
        cout << "enter employee number: \n";
        cin >> e[i].empnum;
        cout << "enter compensation: \n";
        cin >> e[i].sallery;
    }
    cout << "-----EMPLOYEE DATA-----\n";
    for(int i=0;i<3;i++)
    cout << "EMPLOYEE: " << i+1 << ": " << endl << "EMPLOYEE NUMBER : " << e[i].empnum << "employee sallery: " << e[i].sallery << endl;
    return 0;
}
```

```
C:\Users\Taha's Lappy\OneDrive\Documents\lab 3 lab task 6.exe
enter employee number:
1
enter compensation:
23000
enter employee number:
2
enter compensation:
34000
enter employee number:
3
enter compensation:
50000
-----EMPLOYEE DATA-----
EMPLOYEE: 1:
EMPLOYEE NUMBER :1employee sallery: 23000
EMPLOYEE: 2:
EMPLOYEE NUMBER :2employee sallery: 34000
EMPLOYEE: 3:
EMPLOYEE NUMBER :3employee sallery: 50000
-----
Process exited after 10.4 seconds with return value 0
Press any key to continue . . .
```

TASK 8

```
lab task 8.cpp
1 #include<iostream>
2 using namespace std;
3 struct date
4 {
5     int month,day,year;
6 };
7 int main()
8 {
9     date d;
10    char ch;
11    cout << "enter date (MM/DD/YYYY)";
12    cin >> d.month>>ch>>d.day>>ch>>d.year;
13    cout<<"Date is: "<<d.month<<"/"<<d.day<<"/"<<d.year;
14    return 0;
15 }
```

```
enter date (MM/DD/YYYY)02
28
2007
Date is: 2/8/7
-----
Process exited after 12.91 seconds with return value 0
Press any key to continue . . .
```

TASK 9

lab3 lab task 8.cpp [*) lab 3 lab task 6.cpp

```
1 #include<iostream>
2 using namespace std;
3 struct time
4 {
5     int hour,seconds,minutes;
6 };
7 int main()
8 {
9     time t1;
10    char ch;
11    cout << "enter the time (HH:MM:SS)";
12    cin >> t1.hour>>ch>>t1.minutes>>ch>>t1.seconds;
13    long totalsecs = t1.hour*3600 + t1.minutes*60 + t1.seconds;
14
15    cout<<"Total seconds = "<<totalsecs;
16
17    return 0;
18 }
```

C:\Users\Taha's Lappy\OneDrive\Documents\lab3 lab task 8.exe

```
enter the time (HH:MM:SS)2
23
45
Total seconds = 7385
-----
Process exited after 12.29 seconds with return value 0
Press any key to continue . . .
```

HOME TASKSSSS

lab3 lab task 9.cpp labs home task 1.cpp

```
#include <iostream>
using namespace std;
struct hotel
{
    int dishes;
    float bill;
    string name;
};
int main()
{
    hotel h[3];
    float totalbillwithtax=0;
    cout << "enter record of last three hotels: ";
    for(int i=0;i<3;i++)
    {
        cout << "\nHOTEL " << i+1 << endl;
        cout << "hotel name: ";
        cin >> h[i].name;
        cout << "number of dishes eaten: ";
        cin >> h[i].dishes;
        cout << "Total bill(without bill): ";
        cin >> h[i].bill;
        totalbillwithtax+= h[i].bill*1.5;
    }
    cout << "\nTotal bill of three hotels (including 15% tax): " << totalbillwithtax << endl;
    cout << "\n--- Hotel Records ---\n";
    for(int i = 0; i < 3; i++)
    {
        cout << "\nHotel " << i+1 << endl;
        cout << "Name: " << h[i].name << endl;
        cout << "Dishes eaten: " << h[i].dishes << endl;
        cout << "Bill (without tax): " << h[i].bill << endl;
        cout << "Bill (with 15% tax): " << h[i].bill * 1.15 << endl;
    }
    return 0;
}
```

C:\Users\Taha's Lappy\OneDrive\Documents\lab3 home task 1.exe

```
enter record of last three hotels:
HOTEL 1
hotel name: pine hotel
number of dishes eaten: Total bill(without bill):
HOTEL 2
hotel name: number of dishes eaten: Total bill(without bill):
HOTEL 3
hotel name: number of dishes eaten: Total bill(without bill):
Total bill of three hotels (including 15% tax): 0

--- Hotel Records ---

Hotel 1
Name: pine
Dishes eaten: 0
Bill (without tax): 0
Bill (with 15% tax): 0

Hotel 2
Name:
Dishes eaten: 4254288
Bill (without tax): 0
Bill (with 15% tax): 0

Hotel 3
Name:
Dishes eaten: 13570912
Bill (without tax): 0
Bill (with 15% tax): 0
```